

	Internal Exam	Assignm ent	Project Evaluation	End Semester Examinations		
CO 1	✓			✓		
CO 2	✓	✓		✓		
CO 3	✓	✓		✓		
CO 4		✓		✓		
CO 5		✓		✓		
CO 6	✓					
Programme		B. Sc. Statistics				
Course Code		STA2FM106(1)				
Course Title		Managerial Decision Making				
Type of Course		MDC				
Semester		II				
Academic Level		100-199				
Course Details		Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
		3	3	-	-	45
Pre-requisites		HSE level Mathematics Course				
Course Summary		To make students aware of importance of managerial decisions and the use of Statistical theories in developing scientific decisions				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Explain various decision making environments in management	U	C	Instructor-created exams / Quiz/ Seminar presentation
CO2	Discuss the outcome of any payoff	R	F	Practical Assignment / Instructor-created exams
CO3	Assessing the purpose of Inventory for smooth Business operations and critically evaluate ethical implications of statistical methods aligning with human values.	U	C	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO4	Explain the simulation of a real system	U	C	Instructor-created exams / Home Assignments

CO5	Describe the role of game theory in business and analyze data to help entrepreneurial decisions using critical thinking skills	U	F	One Minute Reflection Writing assignments/ Instructor-created exams
CO6	Define probability and discuss expected values.	R	F	Viva Voce/ Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Module	Unit	Content	Hrs (36+9)	Marks (50)
I	Concepts of Decision Making		9	
	1	Environment Uncertainty and Conflict		
	2	Decision Alternatives		
	3	States of Nature		
	4	Pay Off		
	5	Computation of Expected Monetary Value		
	Sections from References:			
II	Inventory		9	
	5	Inventory Management.		
	6	Need and necessity of Inventory		
	7	Parameters of Inventory management.		
	8	Economic Order Quantity with and without lead time		
	Sections from References:			
III	Simulation of Inventory		9	
	9	Simulation		
	10	Monte Carlo Method		
	11	Use of simulation in Inventory		
	12	Game theory		
	13	Strategy, Pay off, Pay off matrix,		
	14	Pure and Mixed strategies, Value of game		
	Sections from References:			
IV	Solving games		9	
	15	Minmax and Maxmin Criteria		
	16	Saddle Point and solution		
	17	Principle of Dominance		
	18	Solving 2x2 games		
	19	Graphical solution of 2xn and nx2 games		
	Sections from References:			
V	Open Ended Module		9	
	1	Basics of Matrices, Scalar and Vector multiplication, Concepts of Probability and Expected Value of Variables		

	Sections from References:		
Books and References: 1. Operations Research V K Kapoor Sultan Chand & Sons 2. Operations Research Kanti Swarup			

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	3	-	2	2	3	3	-	3	-	2	2
CO 2	3	-	-	-	-	-	2	-	-	-	-	2
CO 3	-	-	2	-	-	2	2	2	-	-	3	-
CO 4	-	-	3	-	-	2	2	2	-	-	3	3
CO 5	-	-	3	-	-	2	1	-	-	2	3	2
CO 6	3	2	-	-	-	3	3	3	-	-	-	-

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

6. Quiz / Assignment/ Quiz/ Discussion / Seminar
7. Midterm Exam
8. Programming Assignments (20%)
9. Final Exam (70%)

Mapping of COs to Assessment Rubrics :